AI-Face: A Million-Scale Demographically Annotated AI-Generated Face Dataset and Fairness Benchmark

Anonymous Author(s)

Abstract

We present AI-Face, a million-scale benchmark for evaluating fairness in AI-generated face detection. Existing datasets either emphasize detection utility without demographic annotation or provide demographic labels for only a narrow slice of deepfake data, making it difficult to measure whether detectors remain accurate and equitable across skin tone, gender, age, intersectional groups, post-processing, and domain shift. AI-Face contains 1,646,545 face crops (400,885 real and 1,245,660 fake) from 28 subsets and 37 generation methods spanning deepfake videos, GANs, and diffusion models, together with skin-tone, gender, and age annotations and a standardized evaluation protocol covering utility, group fairness, intersectional fairness, reliability, and cross-domain generalization. We validate the annotation pipeline through human evaluation and benchmark 12 detectors from four detector families. The benchmark shows that high utility does not imply equitable behavior: 10 of 12 detectors exceed 98% AUC in-domain, yet subgroup disparities remain substantial, diffusion subsets induce the largest fairness gaps, post-processing changes fairness non-monotonically, and no detector dominates across CCv2, DF-Platter, and GenData. AI-Face provides a reproducible benchmark for measuring trade-offs among utility, fairness, robustness, and generalization in AI-generated face detection.

AI-Generated

1. Introduction

AI-generated faces now arise from multiple generation pipelines, including face swaps in videos, GAN-based synthesis, and diffusion-based image generation, and their realism has improved rapidly over the last few years [1–4]. This creates a practical need for detectors that are not only accurate, but also reliable across the people depicted in the media and across the transformations that real images undergo after sharing, recompression, or editing.

That need is not well served by current evaluation practice. Prior work has shown demographic disparities in deepfake detection and in face analysis more broadly [5–10]. However, existing AI-generated face datasets typically optimize for scale, generation diversity, or raw utility benchmarking rather than fairness measurement. Some provide demographic labels but only for a limited family of deepfake videos [6]; others cover GANs or diffusion models but do not support demographic fairness evaluation at comparable scale [11, 12]. As a result, current benchmarks give only a partial answer to a basic question: which detectors remain accurate, fair, and robust when both the generator and the depicted population vary?

AI-Face is designed to make that question measurable. The benchmark combines three ingredients that are usually separated: large-scale demographic annotation, broad generative coverage, and a fairness-first evaluation protocol. It contains 1,646,545 face images with skin-tone, gender, and age labels; spans deepfake videos, GANs, and diffusion models; and evaluates 12 representative detectors using utility metrics, group fairness metrics, intersectional fairness metrics, robustness under post-processing, and cross-domain generalization.

Our empirical results show why this combined view matters. In-domain utility is strong, with 10 of 12 detectors above 98% AUC, but fairness rankings are unstable across attributes and settings. PG-FDD achieves the best overall AUC (99.17%) and the best AP (99.69%), while UnivFD yields the best skin-tone F_{MEO} (2.58%), PG-FDD gives the best skin-tone F_{OAE} (0.88%) and intersectional F_{EO} (18.35%), and SPSL attains the highest accuracy (96.35%). Cross-domain testing further shows that no single detector is uniformly best: utility can drop sharply on DF-Platter, while fairness leaders change again on CCv2 and GenData.

Our contributions are fourfold:

- We introduce AI-Face, a million-scale benchmark with 400,885 real and 1,245,660 fake face images from 28 subsets and 37 generation methods, covering deepfake videos, GANs, and diffusion models.
- We provide a benchmark construction pipeline for skin-tone, gender, and age annotation, including a colorimetric skin-tone estimator and a CLIP-based demographic annotator validated with human assessment.
- We define a standardized fairness benchmark that jointly measures utility, group fairness, intersectional fairness, reliability under six post-processings, and cross-domain generalization.

Existing Benchmarks	Year	Category			Scope of Benchmark		
		Deepfake Videos	GAN	DM	Utility	Fairness	
						General	Reliability
Loc et al. [5]	2021	✓			✓	✓	
CCv1 [7]	2021	✓	✓		✓	✓	
DeepfakeBench [14]	2023	✓	✓		✓		
CDDB [17]	2023		✓		✓		
Lin et al. [15]	2024	✓	✓		✓		
Le et al. [16]	2024	✓	✓		✓		
DF40 [4]	2024	✓	✓	✓	✓		
Ours	2025	✓	✓	✓	✓	✓	✓

Table 1. *Comparison of existing AI-generated face detection benchmarks and ours. Fairness 'General' means fairness evaluation under default/basic settings. Fairness 'Reliability' measures fairness consistency across dynamic scenarios (*e.g.*, post-processing).*

- We report benchmark findings that separate utility from fairness, identify subgroup-specific failure modes, and show that fairness robustness and fairness generalization remain unsolved even for strong detectors.

2. Related Work

Benchmarks for AI-generated face detection. Deepfake detection research has been driven by datasets such as FaceForensics++ [1], DFDC [2], DFD [13], and Celeb-DF-v2 [3], with later benchmark efforts standardizing detector implementations and evaluation pipelines [14–16]. More recent datasets extend coverage to additional forgery types or realistic scenarios, including CDDB [17], DF-Platter [12], and DF40 [4]. These resources are valuable for utility benchmarking, but they do not jointly offer million-scale demographic annotation and broad coverage across deepfake, GAN, and diffusion face generation.

Fairness analysis for face and deepfake systems. Demographic performance disparities are well documented in face analysis and biometric systems [8–10]. In deepfake detection, Trinh and Liu [5] and Xu *et al.* [6] show that dataset composition and subgroup imbalance can lead to substantial disparities. Casual Conversations and CCv2 were introduced to study fairness in vision systems under demographic variation [7, 18], but they are evaluation datasets rather than large-scale AI-generated face benchmarks. AI-Face differs by directly targeting fairness evaluation for AI-generated face detection and by combining demographic labels with generation diversity at benchmark scale.

Fairness-aware evaluation and metrics. Fairness in classification has been studied through demographic parity, equalized odds, equality of opportunity, and individual fairness, among other criteria [19–21]. Recent benchmark work emphasizes that fairness conclusions depend strongly on standardized protocols and diverse evaluation settings [22]. AI-Face follows this perspective: the main contribution is an evaluation framework that exposes how detector rankings change across subgroup definitions, intersections, perturbations, and domains.

3. AI-Face Benchmark Design

3.1. Scope and composition

AI-Face contains 1,646,545 cropped face images, including 400,885 real images and 1,245,660 fake images. The real portion is aggregated from FFHQ [23], IMDB-WIKI [24], and the real partitions of FF++ [1], DFDC [2], DFD [13], and Celeb-DF-v2 [3]. The fake portion covers four deepfake video datasets, ten GAN generators, and eight diffusion models, for a total of 37 generation methods. Face crops are extracted and standardized using RetinaFace [25].

This design is motivated by two limitations of prior resources. First, fairness conclusions drawn from only one generation family can be misleading because the artifacts exploited by detectors vary across deepfake videos, GANs, and diffusion images [4, 26, 27]. Second, a benchmark intended for fairness analysis must expose both subgroup diversity and generator diversity. AI-Face explicitly aims to scale both axes at once.

3.2. Demographic annotation

AI-Face annotates skin tone, gender, and age. For skin tone, the construction pipeline first localizes facial landmarks and masked skin regions, then clusters skin pixels with k-means [31], and maps dominant colors to the Monk Skin Tone scale [32]. The released benchmark evaluates skin-tone fairness using three coarse groups (light, medium, dark), while retaining the finer ten-tone annotations for distribution analysis. This choice avoids over-interpreting race labels whose meanings are unstable across datasets and annotation protocols [33].

For gender and age, AI-Face uses a frozen CLIP ViT-L/14 image encoder [34, 35] with a lightweight multilayer classifier trained on IMDB-WIKI [24]. As shown in Figure 1, the annotation training combines an imbalance-aware loss, a fairness regularizer based on Sinkhorn distance [36], and sharpness-aware optimization [37]. We emphasize that this annotator is part of the benchmark construction pipeline: the paper's contribution is the resulting validated benchmark, not a new demographic recognition method.

3.3. Evaluation protocol

AI-Face defines a standard 80/20 train/test split and evaluates detectors along four axes:

1. **Utility:** AUC, ACC, AP, EER, and FPR.
2. **Group fairness:** fairness scores over skin tone, gender, and age. Lower is better for all fairness metrics.
3. **Intersectional fairness:** fairness over the six gender $\times$ skin-tone intersections.
4. **Stress testing:** reliability under six post-processings and cross-domain generalization to CCv2 [18], DF-Platter [12], and GenData [11].

Following the released implementation, AI-Face reports

AI-Generated

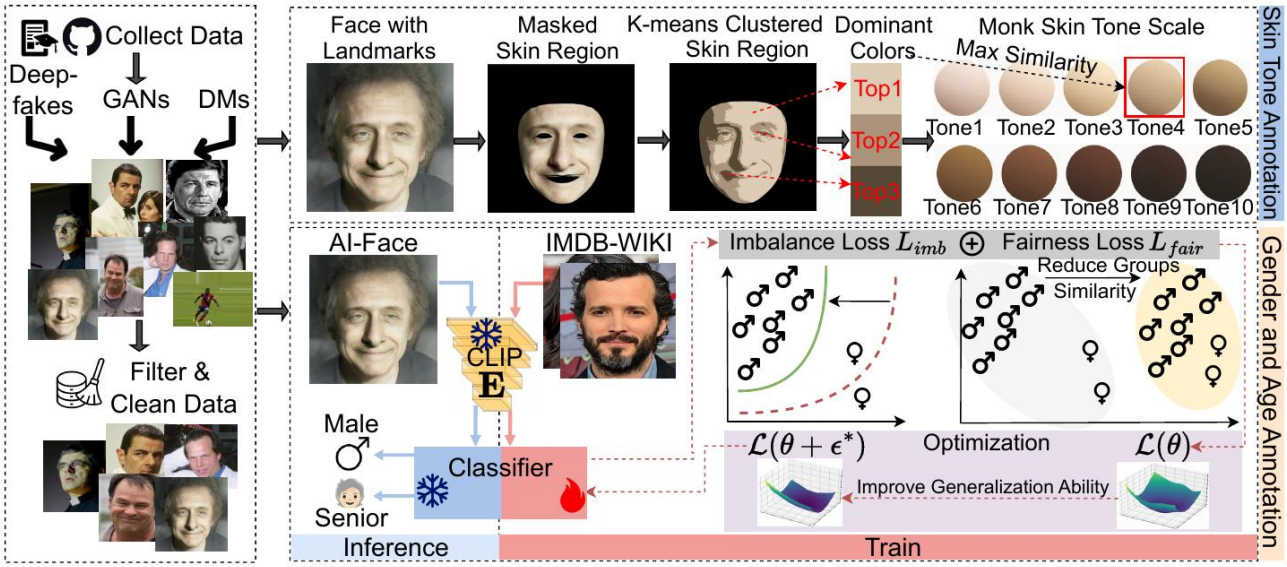

Figure 1. Overview of the AI-Face construction pipeline. AI-Face aggregates real images and AI-generated faces from deepfake, GAN, and diffusion sources; extracts face crops; estimates skin tone by masking skin regions, clustering dominant colors, and mapping to the Monk Skin Tone scale; and predicts gender and age using a frozen CLIP image encoder with a lightweight classifier trained on IMDB-WIKI. The benchmark contribution is the resulting annotated evaluation resource and protocol, not a new detection model.

Dataset	Year	Face Images		Generation Category			#Generation	Source of Real Images	Demographic Annotation		
		#Real	#Fake	Deepfake Videos	GAN	DM	Methods		Skin Tone	Gender	Age
DF-1.0 [28]	2020	2.9M	14.7M	✓			1	Self-Recording		✓	
DeePhy [29]	2022	1K	50.4K	✓	✓		3	YouTube	✓	✓	✓
DF-Platter [12]	2023	392.3K	653.4K	✓	✓		3	YouTube	✓	✓	✓
GenData [11]	2023	-	20K		✓	✓	3	CelebA [30]		✓	
A-FF++ [6]	2024	29.8K	149.1K	✓			5	YouTube		✓	✓
A-DFD [6]	2024	10.8K	89.6K	✓			5	Self-Recording		✓	✓
A-DFDC [6]	2024	54.5K	52.6K	✓	✓		8	Self-Recording		✓	✓
A-Celeb-DF-v2 [6]	2024	26.3K	166.5K	✓			1	Self-Recording		✓	
A-DF-1.0 [6]	2024	870.3K	321.5K	✓			1	Self-Recording		✓	✓
AI-Face	2025	400K	1.2M	✓	✓	✓	37	FFHQ [23], IMDB-WIKI [24], real from FF++ [1], DFDC [2], DFD [13],Celeb-DF-v2 [3]	✓	✓	✓

Table 2. *Quantitative comparison of existing datasets with ours on demographically annotated AI-generated faces.*

F_{DP} for demographic parity disparity, F_{MEO} for the maximum equalized-odds-related disparity across confusion-matrix rates, F_{OAE} for subgroup accuracy disparity, F_{EO} for equalized-odds-style disparity over intersections, and F_{IND} as the benchmark's individual-fairness score. We use the implementation in the released code for all metric computations to keep the protocol reproducible. Reporting these metrics together is important because aggregate AUC can hide systematic subgroup errors when the benchmark is imbalanced. AI-Face therefore reports overall utility and worst-group-style disparities from the same predictions, so that a detector is not credited for high average accuracy while failing badly on smaller demographic slices.

4. Benchmark Validation

4.1. Coverage and distribution

Figure 2 summarizes the benchmark composition. AI-Face is not perfectly balanced, and the benchmark intentionally exposes that fact rather than hiding it. Fake images account for 75.65% of the corpus, with diffusion images comprising the largest fake category (36.19%), followed by deepfake videos (20.04%) and GANs (19.42%). Gender is comparatively balanced (52.73% male, 47.27% female), but age is concentrated in adults (49.67%) and middle-aged faces (26.72%). Skin tones are concentrated in the middle of the Monk scale, especially tones 5 and 6. These skewed marginals are relevant because fairness evaluation should reflect realistic imbalance rather than rely only on artificially balanced subsets.

AI-Generated

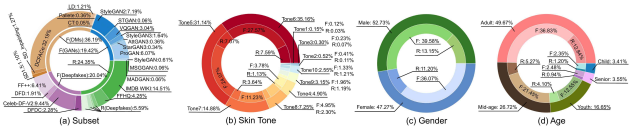

Figure 2. Distribution diagnostics for AI-Face. The benchmark spans real images and three fake generation categories, but demographic marginals remain uneven, especially for age and skin tone. This motivates reporting fairness by subgroup and by intersection instead of only overall accuracy.

Evaluation Type	Dataset	Gender ACC	Gender Precision	Gender Recall	Age ACC	Age Precision	Age Recall
Strategic	A-FF++	8.143	17.583	5.966	37.700	39.459	45.381
	AI-Face	**86.857**	**74.404**	**77.367**	**85.700**	**74.024**	**63.751**
	A-DFDC	21.600	28.604	23.082	33.400	38.011	40.165
	AI-Face	**91.700**	**92.129**	**83.448**	**77.000**	**76.184**	**62.646**
Random	A-Celeb-DF-V2	89.628	90.626	90.494	-		
	AI-Face	**91.206**	**91.474**	**91.767**	-		
	A-DFD	70.900	71.686	74.435	-		
	AI-Face	**92.300**	**91.060**	**91.727**	-		

Table 3. *Annotation quality assessment results (%) for A-FF++, A-DFDC, A-Celeb-DF-V2, A-DFD, and our AI-Face. ACC: Accuracy.*

4.2. Annotation quality

Table 3 reports human validation of AI-Face annotations against prior demographic annotation sets. The study uses three annotators with majority vote and includes both strategically difficult examples and random samples. On strategic samples from A-FF++, AI-Face improves gender accuracy from 8.14% to 86.86% and age accuracy from 37.70% to 85.70%. On strategic A-DFDC samples, gender accuracy rises from 21.60% to 91.70%. On random samples, AI-Face remains consistently stronger, improving over A-DFD by 21.4 points in gender accuracy and slightly improving over A-Celeb-DF-v2 as well. These results do not imply that automatic annotation is error-free, but they do provide concrete evidence that the benchmark labels are substantially more reliable than prior large-scale annotations used for deepfake fairness studies. This validation design is important for benchmark reliability. Strategic samples test whether the annotation pipeline breaks on hard cases that are most likely to distort fairness comparisons, while random samples estimate label quality under the released data distribution. The combined evidence suggests that remaining annotation noise is real but unlikely to dominate the benchmark's main comparative conclusions.

5. Experimental Setup

We benchmark 12 detectors from four families: naive image classifiers (Xception [38], EfficientNet-B4 [39], ViT-B/16 [40]); frequency-based detectors (F3Net [41], SPSL [42], SRM [43]); spatial or generalizable detectors (UCF [44], UnivFD [26], CORE [45]); and fairness-enhanced detectors (DAW-FDD, DAG-FDD [46], and PG-FDD [47]).

We follow the released training code. CNN-based detectors use 256×256 inputs, while ViT-B/16 and UnivFD use 224×224 inputs. Optimization uses SGD with learning rate 5×10^{-4}, momentum 0.9, weight decay 5×10^{-3}, train batch size 128, test batch size 32, and 10 epochs. Reliability experiments apply six post-processings: JPEG compression, Gaussian blur, hue-saturation-value perturbation, brightness/contrast adjustment, rotation, and random crop. Cross-domain fairness generalization is evaluated on the gender attribute because that attribute is shared across the external datasets used in the benchmark; on CCv2, which contains only real images, we report F_{OAE} and ACC.

6. Benchmark Results and Analysis

6.1. In-domain utility and fairness

Table 4 shows that AI-Face is not a trivial benchmark. Utility is high but not saturated: 10 of 12 detectors exceed 98% AUC, yet FPR varies from 10.97% to 20.07%, and fairness disparities remain large. PG-FDD attains the best AUC (99.172%), AP (99.694%), and FPR (10.971%), while SPSL attains the best ACC (96.346%) and EER (4.371%). These strong overall scores could suggest that the task is close to solved if one only looks at utility.

The fairness metrics show a different picture. No detector dominates across all protected attributes or fairness notions. UnivFD has the best skin-tone F_{MEO} (2.577%), PG-FDD has the best skin-tone F_{OAE} (0.882%), Xception has the best gender F_{OAE} (0.975%), EfficientB4 has the best age F_{EO} (10.300%), and PG-FDD has the best intersectional F_{EO} (18.348%). Even the best intersectional gap remains far from negligible. This separation between utility leaders and fairness leaders is the central empirical reason AI-Face is needed as a benchmark rather than just another large detection dataset.

6.2. Intersectional and subset-level behavior

Figure 3 visualizes false-positive rates over the six gender $\times$ skin-tone intersections. A consistent pattern emerges across most detectors: light-skin groups, especially female-light and male-light, tend to have the highest FPR, while male-dark is often the lowest-FPR subgroup. For example, F3Net ranges from 31.88% on female-light to 7.81% on male-dark. PG-FDD reduces this spread substantially but does not eliminate it, ranging from 17.81% to 6.56%.

Figure 4 shows that these disparities are not uniform across generator subsets. Subset difficulty and subset fairness vary substantially, and diffusion-model subsets create the largest fairness excursions for several methods. Stable Diffusion variants and Latent Diffusion frequently induce large spikes in intersectional F_{EO}, even when AUC remains high. This is an important diagnostic property of the benchmark: AI-Face distinguishes between detectors that are strong on classic deepfake videos and detectors that remain stable when the generator family changes.

AI-Generated

Measure	Attribute	Metric	Model Type											
			Naive			Frequency			Spatial			Fairness-enhanced		
			Xception [38]	EfficientB4 [39]	ViT-B/16 [40]	F3Net [41]	SPSL [42]	SRM [43]	UCF [44]	UnivFD [26]	CORE [45]	DAW-FDD [46]	DAG-FDD [46]	PG-FDD [47]
Fairness(%)↓	Skin Tone	F_{MEO}	8.836	8.300	6.264	19.938	8.055	10.002	17.325	**2.577**	10.779	14.118	6.551	6.465
		F_{DP}	9.751	**6.184**	7.728	12.876	9.379	10.897	12.581	8.556	10.317	10.706	8.617	9.746
		F_{OAE}	1.271	4.377	2.168	2.818	1.135	0.915	1.883	2.748	1.332	1.667	1.388	**0.882**
		F_{EO}	12.132	11.062	8.813	23.708	9.789	14.239	21.92	**5.536**	13.069	16.604	7.383	9.115
	Gender	F_{MEO}	3.975	5.385	5.104	4.717	4.411	6.271	5.074	4.503	5.795	5.510	5.910	**3.190**
		F_{DP}	1.691	1.725	1.344	1.864	1.827	1.957	1.736	**1.190**	2.154	2.015	2.151	1.252
		F_{OAE}	**0.975**	1.487	1.803	1.129	1.037	1.772	1.451	1.622	1.389	1.325	1.420	1.071
		F_{EO}	4.143	5.863	6.031	4.870	4.534	6.78	5.510	5.408	5.931	5.696	6.066	**3.702**
	Age	F_{MEO}	27.883	6.796	14.937	38.801	27.614	24.843	47.500	**5.436**	33.882	45.466	15.229	14.804
		F_{DP}	10.905	11.849	11.839	14.906	11.232	11.570	17.049	15.249	12.564	14.106	**9.633**	10.467
		F_{OAE}	7.265	**2.856**	6.838	10.116	7.270	6.524	11.652	3.793	8.760	11.878	5.533	5.009
		F_{EO}	42.216	**10.300**	30.795	55.032	40.943	38.528	67.545	14.148	48.729	64.384	30.182	29.585
	Intersection	F_{MEO}	10.505	17.586	9.384	21.369	10.379	15.142	20.134	**6.119**	15.34	16.565	12.178	9.578
		F_{DP}	14.511	**8.607**	11.535	17.175	13.259	15.186	17.03	14.026	14.301	14.088	11.705	14.697
		F_{OAE}	2.536	8.461	4.928	4.870	**2.464**	3.998	3.536	6.287	2.775	3.547	4.035	3.062
		F_{EO}	24.315	25.114	27.443	47.783	21.679	30.112	43.376	20.255	28.84	33.122	26.295	**18.348**
	Individual	F_{IND}	10.338	25.742	**0.022**	1.872	2.518	7.621	0.767	3.523	0.041	3.772	0.901	0.780
Utility(%)	-	AUC↑	98.583	98.611	98.69	98.714	98.747	97.936	98.082	98.192	98.579	97.811	98.771	**99.172**
		ACC↑	96.308	94.203	94.472	95.719	**96.346**	95.092	95.151	93.651	96.224	95.426	95.722	96.174
		AP↑	99.350	99.542	99.571	99.453	99.356	99.172	99.273	99.400	99.360	99.015	99.498	**99.694**
		EER↓	5.149	6.689	6.372	5.256	**4.371**	6.483	7.708	7.633	5.145	7.063	5.499	4.961
		FPR↓	12.961	20.066	16.426	14.679	13.661	15.746	13.646	18.550	13.410	16.670	14.844	**10.971**
Training Time / Epoch			**1h15min**	2h25min	2h40min	1h18min	1h20min	3h10min	5h05min	4h	1h16min	1h25min	1h17min	7h20min

Table 4. *Overall performance comparison of difference methods on the AI-Face dataset. The best performance is shown in* ***bold****.*

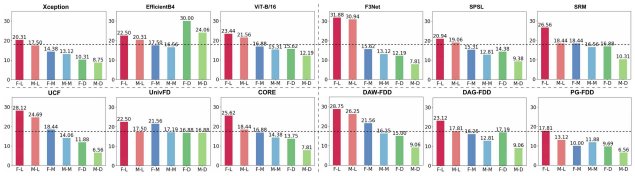

Figure 3. Intersectional false-positive rates on AI-Face. Most detectors show their highest FPR on light-skin groups and their lowest FPR on the male-dark subgroup, indicating that fairness disparities persist even when overall utility is strong.

6.3. Reliability under post-processing

Figure 5 compares post-processing robustness using the performance ratio after versus before each perturbation. Utility ratios stay close to 1 for many methods, but fairness ratios vary much more widely. The spatial family is relatively stable, with UCF, CORE, and UnivFD usually staying close to their original fairness levels. By contrast, fairness-enhanced methods are not uniformly robust: PG-FDD's intersectional F_{EO} ratio rises to roughly 3.5 under rotation, even though its AUC ratio remains below 1 but comparatively stable. This is precisely the kind of measurement gap that AI-Face is intended to expose: a fairness intervention that improves the default setting is not necessarily reliable after common image transformations.

6.4. Effect of training size and real–fake ratio

Figure 6 studies the benchmark itself rather than any single detector. Increasing training size generally improves AUC and often lowers intersectional F_{EO}, but the trend is not monotonic for every method. Likewise, a 1:1 real–fake ratio is often near-optimal for AUC, whereas heavily skewed ratios can amplify fairness gaps. The effect is detector-dependent: SRM becomes markedly less fair at a 10:1 ratio, while some detectors remain comparatively stable. These results support AI-Face's choice to evaluate utility and fairness jointly when analyzing benchmark construction decisions.

6.5. Cross-domain fairness generalization

Table 5 evaluates gender-based fairness generalization on CCv2, DF-Platter, and GenData. The main result is that there is no universal winner. On CCv2, UCF achieves the best ACC (93.009%) and SPSL the best F_{OAE} (0.601%). On DF-Platter, ViT-B/16 yields the best F_{OAE} (0.015%), F3Net the best F_{EO} (6.445%), and PG-FDD the best AUC (85.271%). On GenData, DAG-FDD has the best F_{OAE} (0.714%), EfficientB4 the best F_{EO} (1.995%), and SPSL the best AUC (98.838%).

The magnitude of domain shift is often severe. EfficientB4 loses 30.917 AUC points on DF-Platter, and several other detectors lose more than 20 points. Fairness also changes unpredictably: some detectors become more fair on one target dataset while becoming less fair on another. AI-Face therefore makes a stronger claim than in-domain fairness reporting alone: fairness generalization should be evaluated as a separate property, not inferred from default-setting fairness.

AI-Generated

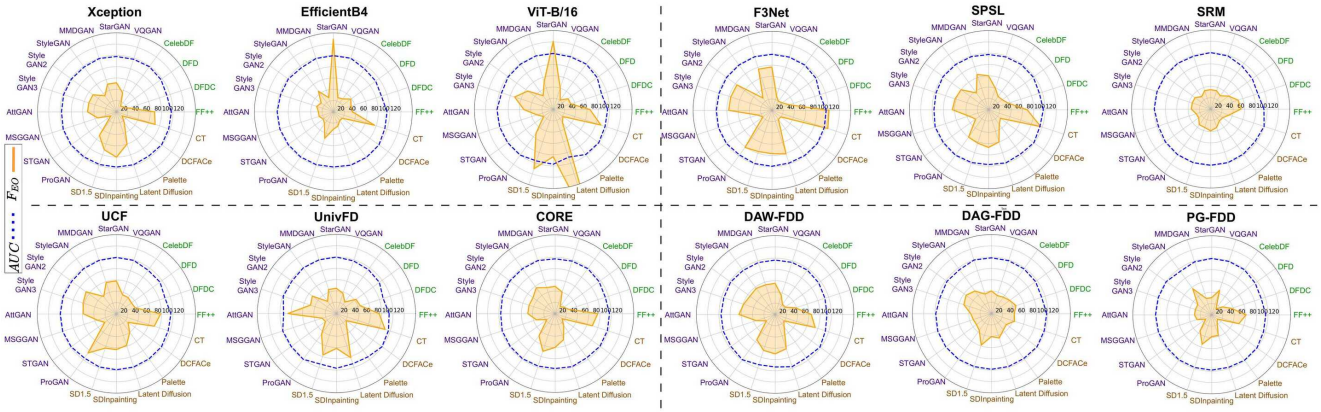

Figure 4. Per-subset visualization of AUC and intersectional F_{EO}. Fairness is much less stable than AUC across subsets, and diffusion-model subsets frequently induce the largest fairness gaps.

Model Type	Detector	Dataset							
		CCv2 [18]		DF-Platter [12]			GenData [11]		
		Fairness(%)↓	Utility(%)↑	Fairness(%)↓		Utility(%)↑	Fairness(%)↓		Utility(%)↑
		F_{OAE}	ACC	F_{OAE}	F_{EO}	AUC	F_{OAE}	F_{EO}	AUC
Naive	Xception	1.006(+0.031)	86.465(-9.843)	6.836(+5.861)	9.789(+5.646)	81.273(-17.310)	2.539(+1.564)	13.487(+9.344)	96.971(-1.612)
	EfficientB4	4.077(+0.259)	82.980(-11.223)	8.786(+7.299)	12.370(+6.507)	67.694(-30.917)	3.304(+1.817)	**1.995** (-3.686)	93.213(-5.398)
	ViT-B/16	2.167(+0.364)	81.489(-12.983)	**0.015** (-1.788)	12.373(+6.342)	76.050(-22.640)	3.164(+1.361)	9.610(+3.579)	88.253(-10.437)
Frequency	F3Net	5.743(+4.614)	87.867(-7.852)	3.521(+2.392)	**6.445** (+1.575)	85.112(-13.602)	1.188(+0.059)	16.306(+11.436)	91.603(-7.111)
	SPSL	**0.601** (-0.436)	80.006(-16.340)	5.109(+4.072)	7.842(+3.308)	82.175(-16.572)	1.385(+0.348)	9.261(+4.272)	**98.838** (+0.091)
	SRM	7.000(+5.228)	79.768(-15.324)	3.823(+2.051)	6.567(-0.213)	66.401(-31.535)	3.281(+1.509)	7.907(+1.127)	90.049(-7.887)
Spatial	UCF	2.169(+0.718)	**93.009** (-2.142)	8.687(+7.236)	17.068(+11.558)	80.821(-17.261)	3.513(+2.062)	10.529(+5.019)	87.778(-10.304)
	UnivFD	7.625(+6.003)	67.983(-25.668)	4.540(+2.918)	9.950(+4.542)	76.443(-21.749)	1.645(+0.023)	3.848(-1.560)	94.418(-3.774)
	CORE	4.410(+3.021)	83.328(-12.896)	7.741(+6.352)	17.348(+11.417)	77.226(-21.353)	3.759(+2.370)	23.289(+17.358)	98.408(-0.171)
Fairness-enhanced	DAW-FDD	4.726(+3.401)	84.685(-10.741)	5.536(+4.211)	13.667(+7.791)	81.807(-16.004)	1.443(+0.118)	10.228(+4.532)	97.854(+0.043)
	DAG-FDD	2.364(+0.944)	83.918(-11.804)	3.064(+1.644)	22.203(+16.137)	75.206(-23.565)	**0.714** (-0.706)	10.332(+4.266)	92.108(-6.663)
	PG-FDD	1.513(+0.442)	92.852(-3.322)	4.565(+3.494)	9.717(+6.015)	**85.271** (-13.901)	3.063(+1.992)	9.479(+5.777)	93.329(-5.843)

Table 5. *Fairness generalization results based on the gender attribute. The smallest performance changes (in parentheses) and the best performance are in red and in* ***bold****, respectively. Only F_{OAE} fairness metric and ACC metric are used in CCv2 due to all samples are real.*

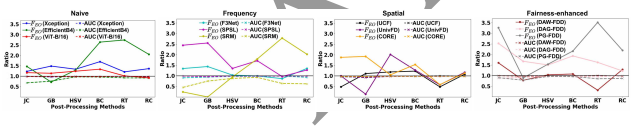

Figure 5. Reliability under six post-processings: JPEG compression (JC), Gaussian blur (GB), hue-saturation-value shift (HSV), brightness/contrast (BC), rotation (RT), and random crop (RC). Solid lines show the ratio of intersectional F_{EO} after versus before perturbation; dashed lines show the corresponding AUC ratio. Fairness changes are often larger and less stable than utility changes.

AI-Generated

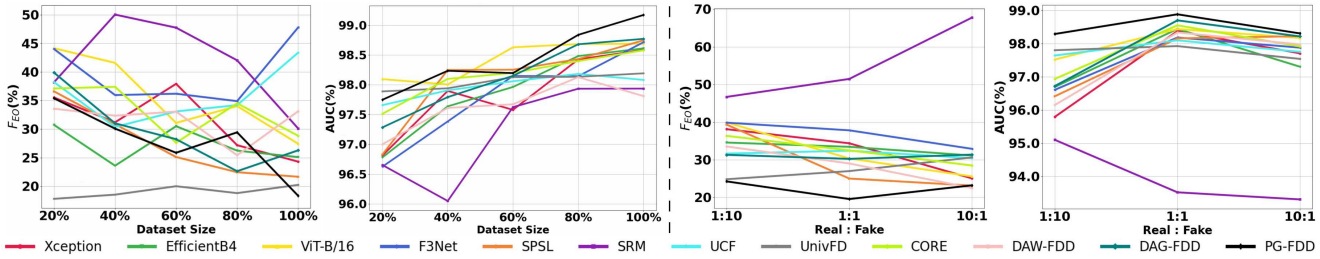

Figure 6. Benchmark sensitivity to dataset scale and real–fake composition. Larger training sets generally help utility and often help fairness, but neither trend is monotonic across all detectors. Skewed real–fake ratios can preserve AUC while substantially changing intersectional fairness.

AI-Generated

7. Discussion

AI-Face contributes a benchmark rather than a new detection algorithm, and its value lies in what it enables researchers to measure. First, it shows that strong in-domain utility does not imply equitable behavior across demographic groups. Second, it reveals that fairness depends on generator family, with diffusion subsets frequently exposing the largest disparities. Third, it makes clear that robustness and fairness generalization are distinct evaluation dimensions: a detector can look fair on the default split yet become much less fair after routine image processing or after transfer to a different domain.

At the same time, AI-Face does not solve fairness evaluation in full. Most demographic labels are automatically inferred, although human validation indicates that the annotations are substantially more reliable than previous large-scale alternatives. The benchmark is also not demographically balanced; age and skin-tone marginals remain skewed, and these skews should be considered when interpreting results. In addition, AI-Face measures fairness for cropped-face binary detection rather than full audiovisual pipelines or broader social harms. High performance on AI-Face should therefore be interpreted as evidence about this evaluation setting, not as a guarantee of fairness in all deployment contexts. For benchmark users, a practical implication is that detector reports should include subgroup, intersectional, and cross-domain slices by default. Headline utility alone is often insufficient and can invert the apparent ranking of methods once fairness reliability is taken into account.

8. Conclusion

AI-Face introduces a million-scale, demographically annotated benchmark for AI-generated face detection that jointly evaluates utility, group fairness, intersectional fairness, reliability, and cross-domain generalization. The benchmark reveals that current detectors achieve high in-domain accuracy while still exhibiting substantial and setting-dependent demographic disparities. By making these trade-offs measurable across deepfake, GAN, and diffusion faces, AI-Face provides a stronger basis for evaluating progress toward fairer and more reliable media forensics systems.

AI-Generated

References

[1] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "Faceforensics++: Learning to detect manipulated facial images," in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 1–11, 2019.

[2] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer, "The deepfake detection challenge (dfdc) dataset," *arXiv preprint arXiv:2006.07397*, 2020.

[3] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, "Celeb-df: A large-scale challenging dataset for deepfake forensics," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 3207–3216, 2020.

[4] Z. Yan, T. Yao, S. Chen, Y. Zhao, X. Fu, J. Zhu, D. Luo, L. Yuan, C. Wang, S. Ding, *et al.*, "Df40: Toward next-generation deepfake detection," *NeurIPS*, 2024.

[5] L. Trinh and Y. Liu, "An examination of fairness of ai models for deepfake detection," *IJCAI*, 2021.

[6] Y. Xu, P. Terhöst, M. Pedersen, and K. Raja, "Analyzing fairness in deepfake detection with massively annotated databases," *IEEE Transactions on Technology and Society*, 2024.

[7] C. Hazirbas, J. Bitton, B. Dolhansky, J. Pan, A. Gordo, and C. C. Ferrer, "Towards measuring fairness in ai: the casual conversations dataset," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 4, no. 3, pp. 324–332, 2021.

[8] C. M. Cook, J. J. Howard, Y. B. Sirotin, J. L. Tipton, and A. R. Vemury, "Demographic effects in facial recognition and their dependence on image acquisition: An evaluation of eleven commercial systems," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 1, no. 1, pp. 32–41, 2019.

[9] V. Albiero, K. Ks, K. Vangara, K. Zhang, M. C. King, and K. W. Bowyer, "Analysis of gender inequality in face recognition accuracy," in *Proceedings of the ieee/cvf winter conference on applications of computer vision workshops*, pp. 81–89, 2020.

[10] K. Krishnapriya, V. Albiero, K. Vangara, M. C. King, and K. W. Bowyer, "Issues related to face recognition accuracy varying based on race and skin tone," *IEEE Transactions on Technology and Society*, vol. 1, no. 1, pp. 8–20, 2020.

[11] C. Teo, M. Abdollahzadeh, and N.-M. M. Cheung, "On measuring fairness in generative models," *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[12] K. Narayan, H. Agarwal, K. Thakral, S. Mittal, M. Vatsa, and R. Singh, "Df-platter: multi-face heterogeneous deepfake dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9739–9748, 2023.

[13] G. Research, "Contributing data to deepfake detection research," 2019. Accessed: 2024-04-12.

[14] Z. Yan, Y. Zhang, X. Yuan, S. Lyu, and B. Wu, "Deepfakebench: A comprehensive benchmark of deepfake detection," in *NeurIPS*, 2023.

[15] J. Deng, C. Lin, P. Hu, C. Shen, Q. Wang, Q. Li, and Q. Li, "Towards benchmarking and evaluating deepfake detection," *IEEE Transactions on Dependable and Secure Computing*, 2024.

[16] B. M. Le, J. Kim, S. Tariq, K. Moore, A. Abuadbba, and S. S. Woo, "Sok: Facial deepfake detectors," *arXiv*, 2024.

[17] C. Li *et al.*, "A continual deepfake detection benchmark: Dataset, methods, and essentials," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 1339–1349, 2023.

[18] B. Porgali, V. Albiero, J. Ryda, C. C. Ferrer, and C. Hazirbas, "The casual conversations v2 dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10–17, 2023.

[19] C. Dwork, M. Hardt, T. Pitassi, O. Reingold, and R. Zemel, "Fairness through awareness," in *Proceedings of the 3rd innovations in theoretical computer science conference*, pp. 214–226, 2012.

[20] R. Williamson and A. Menon, "Fairness risk measures," in *International conference on machine learning*, pp. 6786–6797, PMLR, 2019.

[21] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, "A survey on bias and fairness in machine learning," *ACM computing surveys (CSUR)*, vol. 54, no. 6, pp. 1–35, 2021.

[22] X. Han, J. Chi, Y. Chen, Q. Wang, H. Zhao, N. Zou, and X. Hu, "Ffb: A fair fairness benchmark for in-processing group fairness methods," in *ICLR*, 2024.

[23] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4401–4410, 2019.

[24] R. Rothe, R. Timofte, and L. Van Gool, "Dex: Deep expectation of apparent age from a single image," in *Proceedings of the IEEE international conference on computer vision workshops*, pp. 10–15, 2015.

[25] J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, "Retinaface: Single-shot multi-level face localisation in the wild," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 5203–5212, 2020.

[26] U. Ojha, Y. Li, and Y. J. Lee, "Towards universal fake image detectors that generalize across generative models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 24480–24489, 2023.

AI-Generated

[27] R. Corvi, D. Cozzolino, G. Zingarini, G. Poggi, K. Nagano, and L. Verdoliva, "On the detection of synthetic images generated by diffusion models," in *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, IEEE, 2023.

[28] L. Jiang, R. Li, W. Wu, C. Qian, and C. C. Loy, "Deeperforensics-1.0: A large-scale dataset for real-world face forgery detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 2889–2898, 2020.

[29] K. Narayan, H. Agarwal, K. Thakral, S. Mittal, M. Vatsa, and R. Singh, "Deephy: On deepfake phylogeny," in *2022 IEEE International Joint Conference on Biometrics (IJCB)*, pp. 1–10, IEEE, 2022.

[30] Z. Liu, P. Luo, X. Wang, and X. Tang, "Deep learning face attributes in the wild," in *Proceedings of International Conference on Computer Vision (ICCV)*, December 2015.

[31] J. A. Hartigan and M. A. Wong, "Algorithm as 136: A k-means clustering algorithm," *Journal of the royal statistical society. series c (applied statistics)*, vol. 28, no. 1, pp. 100–108, 1979.

[32] Google, "The Monk Skin Tone Scale," 2024. [Accessed October 16, 2024].

[33] Z. Khan and Y. Fu, "One label, one billion faces: Usage and consistency of racial categories in computer vision," in *Proceedings of the 2021 acm conference on fairness, accountability, and transparency*, pp. 587–597, 2021.

[34] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*, pp. 8748–8763, PMLR, 2021.

[35] G. Ilharco, M. Wortsman, R. Wightman, C. Gordon, N. Carlini, R. Taori, A. Dave, V. Shankar, H. Namkoong, J. Miller, H. Hajishirzi, A. Farhadi, and L. Schmidt, "Open clip," https://github.com/mlfoundations/open_clip, 2021.

[36] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transport," *Advances in neural information processing systems*, vol. 26, 2013.

[37] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur, "Sharpness-aware minimization for efficiently improving generalization," in *International Conference on Learning Representations*, 2020.

[38] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1251–1258, 2017.

[39] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*, pp. 6105–6114, PMLR, 2019.

[40] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *9th International Conference on Learning Representations*, 2021.

[41] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, "Thinking in frequency: Face forgery detection by mining frequency-aware clues," in *European conference on computer vision*, pp. 86–103, Springer, 2020.

[42] H. Liu, X. Li, W. Zhou, Y. Chen, Y. He, H. Xue, W. Zhang, and N. Yu, "Spatial-phase shallow learning: rethinking face forgery detection in frequency domain," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 772–781, 2021.

[43] Y. Luo, Y. Zhang, J. Yan, and W. Liu, "Generalizing face forgery detection with high-frequency features," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 16317–16326, 2021.

[44] Z. Yan, Y. Zhang, Y. Fan, and B. Wu, "Ucf: Uncovering common features for generalizable deepfake detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 22412–22423, 2023.

[45] Y. Ni, D. Meng, C. Yu, C. Quan, D. Ren, and Y. Zhao, "Core: Consistent representation learning for face forgery detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12–21, 2022.

[46] Y. Ju, S. Hu, S. Jia, G. H. Chen, and S. Lyu, "Improving fairness in deepfake detection," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 4655–4665, 2024.

[47] L. Lin, X. He, Y. Ju, X. Wang, F. Ding, and S. Hu, "Preserving fairness generalization in deepfake detection," *CVPR*, 2024.

AI-Generated